

— INSIGHT · AI INFRASTRUCTURE

AI DATA CENTER INFRASTRUCTURE PLANNING: START WITH A WHOLE-SYSTEM BASIS

AI infrastructure planning should begin with an approved set of compute and deployment assumptions, then follow those requirements through power, thermal systems, structure, space, controls, commissioning, operations and future change.

JLL, Uptime Institute and the ASHRAE/PNNL/NEMA AI Data Center Energy Performance Framework all connect AI infrastructure to multiple facility systems. The lesson is practical: treating AI as a cooling-only or rack-only problem leaves major dependencies unresolved.

DEFINE WHAT IS KNOWN ABOUT THE WORKLOAD

The team should record:

- The compute platform and configuration being evaluated.
- The initial load basis and expected deployment sequence.
- Which values are vendor requirements, owner requirements or planning assumptions.
- How much change the design must accommodate.
- What availability, redundancy and operating requirements apply.
- What data is missing and who owns it.

Do not allow a provisional planning value to become an implied facility guarantee.

COORDINATE POWER AND THERMAL DECISIONS

Electrical and thermal requirements affect one another. Load assumptions influence distribution, generation, space and controls. Thermal choices affect power use, water considerations, equipment arrangement, structure, maintenance and commissioning.

The right strategy depends on the actual equipment and operating basis. No single cooling architecture or rack density applies to every AI program.

EXAMINE THE PHYSICAL AND OPERATING INTERFACES

AI planning should address more than capacity totals:

1. Equipment dimensions, weights, access and replacement paths.
2. Electrical and thermal connection points.
3. Controls, alarms and data ownership.
4. Installation and commissioning sequence.
5. Maintenance access and isolation needs.
6. Expansion or technology-change assumptions.
7. The boundary between facility systems and compute equipment.

These interfaces belong in the design basis and change-control process.

PLAN COMMISSIONING AROUND THE OPERATING REQUIREMENT

Commissioning criteria should trace back to the project basis. The team needs to know what evidence demonstrates readiness, which systems must perform together, what prerequisites apply and how issues are closed before acceptance.

CONSIDER MAINTAINABILITY BEFORE TURNOVER

The operating team needs access, documentation, system ownership and a clear route for alarms, findings and change. Decisions made during design can either support or complicate that work.

WHERE EVOLVE FITS

AI Data Centers are part of Evolve's documented Design & Build practice. Evolve supports planning, design, procurement, construction, commissioning, power, monitoring, maintenance and long-term support.

EXPLORE AI DATA CENTERS

EXPLORE POWER GENERATION

DIRECT ANSWERS

WHAT SHOULD BE CONFIRMED FIRST FOR AN AI DATA CENTER?

Confirm the compute and deployment assumptions, their source, their approval state and how they translate into facility requirements.

IS AI INFRASTRUCTURE MAINLY A COOLING PROBLEM?

No. It is a whole-system problem involving electrical, thermal, structural, spatial, controls, commissioning and operating decisions.

HOW SHOULD TEAMS HANDLE CHANGING TECHNOLOGY?

Record assumptions, define acceptable ranges where supported, preserve clear interfaces and evaluate changes against the defined system basis.

Prepared by Evolve Data Center Solutions. Market sources retrieved September 7, 2026. Market facts are attributed to their sources and are not Evolve performance claims.

SOURCES

- [JLL, 2026 Global Data Center Outlook](#) (January 2026)
- [Uptime Institute, Global Data Center Survey 2026](#) (July 2026)
- [ASHRAE / PNNL / NEMA, AI Data Center Energy Performance Framework](#) (retrieved September 7, 2026)

